(i) **To solve an equation of the form** $\sin\theta = k \ \ (-1 \leq k \leq 1)$:

Let α be the numerically smallest angle such that $\sin\alpha = k$. Thus,

$$\begin{aligned} \sin\theta &= \sin\alpha \\ \sin\theta - \sin\alpha &= 0 \\ 2\cos\frac{\theta+\alpha}{2}\sin\frac{\theta-\alpha}{2} &= 0 \ \text{, which gives either} \\ \cos\frac{\theta+\alpha}{2} = 0 \ \text{ or } \ \sin\frac{\theta-\alpha}{2} &= 0. \end{aligned}$$

Now, $\cos\dfrac{\theta+\alpha}{2} = 0$	Now, $\sin\dfrac{\theta-\alpha}{2} = 0$,
$\dfrac{\theta+\alpha}{2} = (2n+1)\dfrac{\pi}{2}, n \in \mathbb{Z}$	$\dfrac{\theta-\alpha}{2} = n\pi, n \in \mathbb{Z}$
$\theta = (2n+1)\,\pi - \alpha, n \in \mathbb{Z}$ ...(i)	$\theta = 2n\pi + \alpha, n \in \mathbb{Z}$ ...(ii)

Combining (i) and (ii), we have $\sin\theta = \sin\alpha \Rightarrow \theta = n\pi + (-1)^n\,\alpha, n \in \mathbb{Z}$. (3.13)

(ii) **To solve an equation of the form** $\cos\theta = k \ \ (-1 \leq k \leq 1)$:

Let α be the numerically smallest angle such that $\cos\alpha = k$. Thus,

$$\begin{aligned} \cos\theta &= \cos\alpha \\ \cos\theta - \cos\alpha &= 0 \\ 2\sin\frac{\theta+\alpha}{2}\sin\frac{\alpha-\theta}{2} &= 0 \\ \sin\frac{\theta+\alpha}{2}\sin\frac{\theta-\alpha}{2} &= 0 \ \text{ , which gives either} \\ \sin\frac{\theta+\alpha}{2} = 0 \ \text{ or } \ \sin\frac{\theta-\alpha}{2} &= 0 \end{aligned}$$

$\sin\dfrac{\theta+\alpha}{2} = 0$ gives	$\sin\dfrac{\theta-\alpha}{2} = 0$ gives
$\dfrac{\theta+\alpha}{2} = n\pi, n \in \mathbb{Z}$	$\dfrac{\theta-\alpha}{2} = n\pi, n \in \mathbb{Z}$
$\theta = 2n\pi - \alpha, n \in \mathbb{Z}$. ...(i)	$\theta = 2n\pi + \alpha, n \in \mathbb{Z}$ ...(ii)

Combining (i) and (ii), we have $\cos\theta = \cos\alpha \Rightarrow \theta = 2n\pi \pm \alpha, n \in \mathbb{Z}$. (3.14)

(iii) **To solve an equation of the form** $\tan\theta = k \ \ (-\infty < k < \infty)$:

Let α be the numerically smallest angle such that $\tan\alpha = k$. Thus,

$$\begin{aligned} \tan\theta &= \tan\alpha \\ \frac{\sin\theta}{\cos\theta} &= \frac{\sin\alpha}{\cos\alpha} \Rightarrow \sin\theta\cos\alpha - \cos\theta\sin\alpha = 0 \\ \sin(\theta-\alpha) &= 0 \Rightarrow \theta - \alpha = n\pi \\ \theta &= n\pi + \alpha, n \in \mathbb{Z} \\ \text{Thus, } \ \tan\theta &= \tan\alpha \Rightarrow \theta = n\pi + \alpha, n \in \mathbb{Z} \end{aligned}$$ (3.15)